

Gibbs-Based Information Criteria and the Over-Parameterized Regime

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Preliminaries

Let $S = \{Z_i\}_{i=1}^n$, where $Z_i = \{(X_i, Y_i)\} \in \mathcal{Z}$ i.i.d. from the data distribution P_Z with realization $z^n = (x^n, y^n)$

Parameter for a machine learning model $w \in \mathcal{W}$

Performance of model is measured by loss function

$L : \mathcal{W} \times \mathcal{Z} \rightarrow \mathbb{R}$ and the log loss $L_{\log}(w, z) = -\log P(y|x; w)$

Define empirical risk and the population risk:

$$L_E(w, z^n) = \frac{1}{n} \sum_{i=1}^n L(w, z_i)$$

$$L_P(w, P_Z) = \mathbb{E}_{P_Z}[L(w, Z)]$$

Here we can just take

$$\hat{W}_{ML} = \operatorname{argmax}_w \hat{L}(w) = \operatorname{argmax}_w \sum_{i=1}^n \log P_k(y_i|x_i; w)$$

$$\overline{\text{gen}}(P_{\hat{W}|S}, P_S) = \mathbb{E}_{P_{\hat{W}, S}}[L_P(\hat{W}, P_Z) - L_E(\hat{W}, S)]$$

Here $P_{\hat{W}, S} = P_{\hat{W}|S} \otimes P_S$

Classical forms of AIC and BIC

K candidate models $M_1, M_2, \dots, M_K$

Each M_k defined by $P_k(y|x; \theta_k)$ and $\pi_k(\theta_k)$, where $\theta_k \in \mathcal{W}_k \in \mathbb{R}^{p_k}$

AIC

Minimize L_P based on D_{KL} (from now just D) between P_Z and parametric distribution

$$\operatorname{argmin}_k D(P_Z || P_k(y|x; \hat{\theta}_{ML}^{(k)})) = \operatorname{argmin}_k \mathbb{E}_{P_Z}[-\log P_k(y|x; \hat{\theta}_{ML}^{(k)})] = \operatorname{argmin}_k L_P(\hat{\theta}_{ML}^{(k)}, P_Z); \text{ AIC} = -\frac{\hat{L}(\hat{\theta}_{ML})}{n} + \frac{p}{n}$$

BIC

Minimize L_E approximating the marginal likelihood of observing z^n from M_k i.e. $m_k(z^n) = \int P_k(y^n|x^n; \theta_k) \pi_k(\theta_k) d\theta_k$

Laplace gives $\log m_k(z^n) = \hat{L}_k(\hat{\theta}_{ML}^{(k)}) - \frac{p_k}{2} \log n + O(1), n \rightarrow \infty$

$$\text{BIC} = -\frac{\hat{L}(\hat{\theta}_{ML})}{n} + \frac{p \log n}{2n}$$

Information Risk Minimization and Gibbs Algorithm

Lemma 1

Suppose $L(w, z) \in [0, 1]$ is bounded, and $S = \{Z_i\}_{i=1}^n$ contains n i.i.d. training samples, then $|\overline{gen}(P_{\hat{W}|S}, P_S)| \leq \sqrt{I(\hat{W}; S)/(2n)}$

This brings us $P_{\hat{W}|S}^* = \operatorname{argmin}_{P_{\hat{W}|S}} \mathbb{E}_{P_{\hat{W},S}}[L_E(\hat{W}, S)] + \frac{1}{\beta} I(\hat{W}; S)$
with $D(P_{\hat{W}|S} || \pi | P_S) \geq I(\hat{W}; S)$

Lemma 2

$$\min_{P_{\hat{W}|S}} \mathbb{E}_{P_{\hat{W},S}}[L_E(\hat{W}, S)] + \frac{1}{\beta} D(P_{\hat{W}|S} || \pi | P_S) = \\ -\frac{1}{\beta} \mathbb{E}_{P_S}[\log \mathbb{E}_{\pi}[e^{-\beta L_E(W, S)}]]$$

which can be achieved by Gibbs algorithm

$$P_{\hat{W}|S}^*(w|s) = \frac{\pi(w)e^{-\beta L_E(w,s)}}{\mathbb{E}_{\pi}[e^{-\beta L_E(W,s)}]} \text{ for } \beta > 0$$

Sampling Algorithms for Gibbs distribution

It is hard to compute $\mathbb{E}_\pi[e^{-\beta L_E(W,s)}]$ directly

SGLD

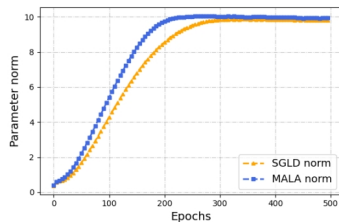
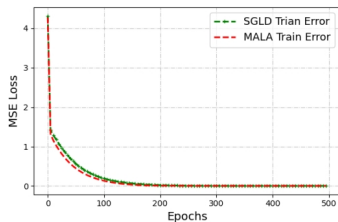
$$\hat{W}_{t+1} = \hat{W}_t - \eta \nabla L_E(\hat{W}_t, s) + \sqrt{\frac{2\eta}{\beta}} \xi_t \text{ where } \xi_t \sim \mathcal{N}(0, I)$$

It was shown that under some conditions Wasserstein distance between induced distribution and Gibbs distribution will converge to zero as $t \rightarrow \infty$

MALA

Metropolis-adjusted Langevin algorithm also is first-order sampling method. Difference in additional Metropolis-adjusted step, which provides a faster convergence rate

Experimental Results



Gibbs-based AIC

Proposition 1

For the Gibbs algorithm $P_{\hat{W}|S}^*$, the expected generalization error is:

$$\overline{gen}(P_{\hat{W}|S}^*, P_S) = \frac{I_{\text{SKL}}(P_{\hat{W}|S}^*, P_S)}{\beta}$$

$$I_{\text{SKL}}(P_{Y|X}, P_X) \triangleq D(P_{X,Y} \| P_X \otimes P_Y) + D(P_X \otimes P_Y \| P_{X,Y})$$

This leads to the **Gibbs-based AIC**:

$$\text{AIC}^+ \triangleq L_E(\hat{W}_{\text{Gibbs}}, z^n) + \frac{1}{\beta} I_{\text{SKL}}(P_{\hat{W}|S}^*, P_S)$$

Theorem 1 (Classical Regime)

$\beta = n$, p is fixed and $n \rightarrow \infty$:

$$\text{AIC}^+ = L_E(\hat{W}_{\text{Gibbs}}, z^n) + \frac{p}{n}$$

Gibbs-based BICs

Proposition 2

For the Gibbs algorithm $P_{\hat{W}|S}^*$ with log-loss and $\beta = n$, the negative log-marginal likelihood is:

$$-\frac{1}{n} \log m(z^n) = \mathbb{E}_{P_{\hat{W}|S=z^n}^*} [L_E(\hat{W}, z^n)] + \frac{1}{n} D(P_{\hat{W}|S=z^n}^* \| \pi)$$

This motivates two versions of **Gibbs-based BIC**:

$$\begin{aligned} \text{BIC}^+ &\triangleq L_E(\hat{W}_{\text{Gibbs}}, z^n) + \frac{1}{n} D(P_{\hat{W}|S=z^n}^* \| \pi) \\ \text{BIC}^- &\triangleq \mathbb{E}_{\pi} [L_E(W, z^n)] - \frac{1}{n} D(\pi \| P_{\hat{W}|S=z^n}^*) \end{aligned}$$

Theorem 2 (Classical Regime)

For p fixed and $n \rightarrow \infty$:

$$\text{BIC}^+ = L_E(\hat{W}_{\text{Gibbs}}, z^n) + \frac{p \log n}{2n}$$

Over-Parameterized Regime and Random Feature Model

Key Problem

Classical AIC/BIC fail in over-parameterized regime ($p > n$) due to:

- ▶ Asymptotic normality of MLE breaks down
- ▶ Laplace approximation invalid
- ▶ Infinite number of interpolating solutions

Random Feature (RF) Model

Two-layer neural network with fixed first layer:

$$g(x) = \sum_{j=1}^p f\left(\frac{\langle x, F_j \rangle}{\sqrt{d}}\right) w_j = f\left(\frac{x^\top F}{\sqrt{d}}\right) w$$

where F is random feature matrix with i.i.d. $\mathcal{N}(0, 1)$ entries.

Gibbs posterior for RF model is Gaussian: $P_{w|S}^* \sim \mathcal{N}(\hat{W}_\lambda, \Sigma_w)$

Over-Parameterized Gibbs-Based BICs

For RF model, KL divergences have closed forms:

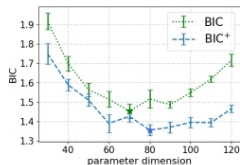
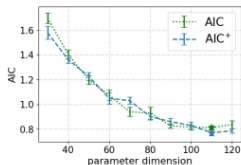
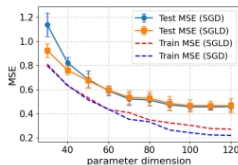
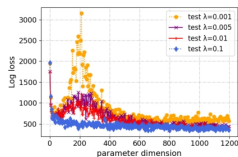
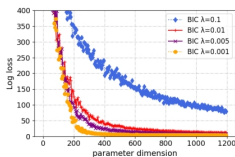
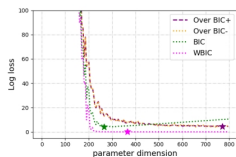
BIC⁺ for RF Model

$$\text{BIC}^+ \triangleq L_E(\hat{W}_{\text{Gibbs}}, z^n) + \underbrace{\frac{\lambda}{2\sigma^2} \|\hat{W}_\lambda\|_2^2}_{\ell_2 \text{ term}} + \underbrace{\frac{1}{2} V(1/\lambda, r) - \frac{\lambda}{8} \mathcal{F}(1/\lambda, r)}_{\text{covariance term}}$$

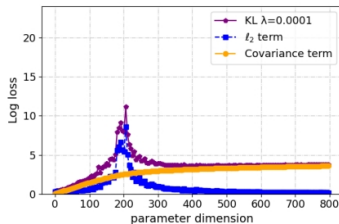
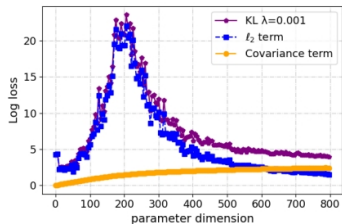
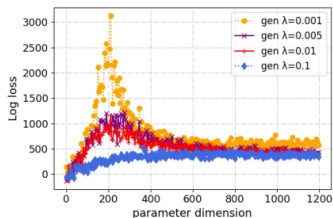
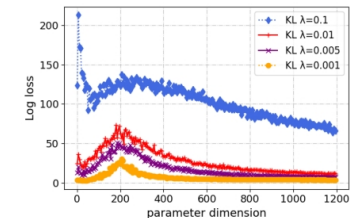
where $r = p/n$, and $\mathcal{F}$, V are defined via random matrix theory.

$$\text{BIC}^- \triangleq \mathbb{E}_\pi[L_E(W, z^n)] - \frac{1}{2n} \hat{W}_\lambda^\top (\Sigma_w)^{-1} \hat{W}_\lambda - \frac{1}{2n} \text{tr}(I_p + \frac{1}{\lambda n} B^\top B) + \frac{1}{2} V(1/\lambda, r) + \frac{p}{2n}$$

Experimental Results



Experimental Results



Key Findings

Mismatch Between BIC and Population Risk

- ▶ BIC^+ does **not** exhibit double-descent behavior
- ▶ Marginal likelihood (BIC) and generalization error (AIC) behave differently
- ▶ Prior distribution (λ) significantly affects this mismatch

Similarity Between KL Divergence and Generalization

- ▶ For fixed λ , KL divergence and I_{SKL} show similar trends
- ▶ ℓ_2 norm of weights decreases with over-parameterization
- ▶ Smaller ℓ_2 norm correlates with better generalization

Conclusion

Contributions

1. Developed Gibbs-based AIC/BIC using information-theoretic analysis
2. Extended analysis to over-parameterized regime for RF models
3. Provided exact characterization of marginal likelihood via information measures
4. Demonstrated mismatch between marginal likelihood and generalization error

Implications

- ▶ Gibbs-based criteria provide unified framework for classical and over-parameterized regimes
- ▶ Model selection requires careful consideration of prior in over-parameterized setting
- ▶ Information measures offer insights into double-descent phenomena